



Solution to an equation

Task A: Testing solutions to equations

- 1) Fill in the table for both expressions. Use your table to find the **solution** to the equation.

a)	equation		$x = 1$	$x = 2$	$x = 3$	$x = 4$	$x = 5$	solution
	$2x + 3 = 4x - 5$	$2x + 3$	5	7	9	11	13	$x = 4$
		$4x - 5$	-1	3	7	11	15	

b)	equation		$x = 2$	$x = 0$	$x = -2$	$x = -4$	$x = -6$	solution
	$3(x - 4) = 5x$	$3(x - 4)$	-6	-12	-18	-24	-30	$x = -6$
		$5x$	10	0	-10	-20	-30	

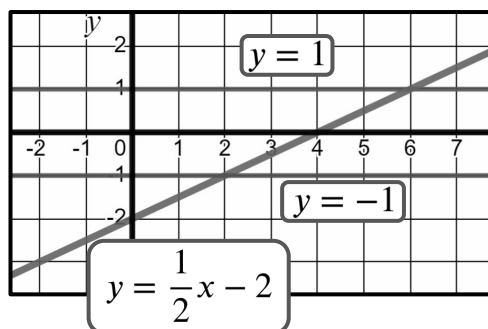
c)	equation		$x = 1$	$x = \frac{3}{2}$	$x = 2$	$x = \frac{5}{2}$	$x = 3$	solution
	$20 - 2x = 2x + 10$	$20 - 2x$	18	17	16	15	14	$x = \frac{5}{2}$
		$2x + 10$	12	13	14	15	16	

- 2) Sort these equations into the correct column in the table.

has $x = 2$ as a solution	has $x = 0$ as a solution	has no solution
$5x = 10$ $5(x + 1) = 15$ $3x - 6 = 0$ $2x - 2 = 6 - 2x$	$x = 3x$ $x - 2 = 10x - 2$	$2x + 5 = 2x + 3$ $2 - x = 7 - x$ $3(2x + 5) = 6x - 2$

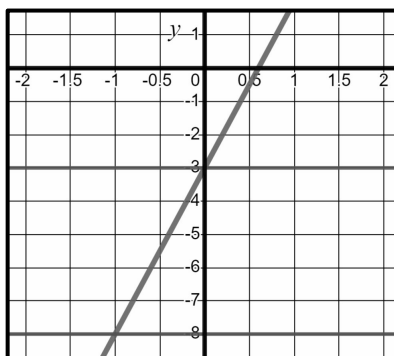
Task B: Finding solutions from graphs

1a) Use the lines on the graph to solve the equations below.



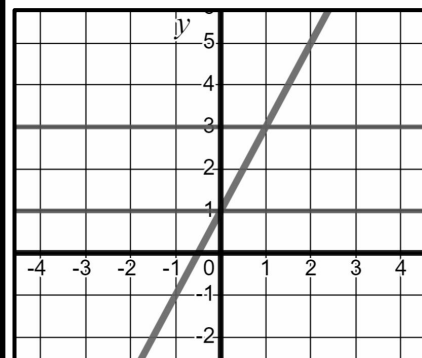
equation	solution
$\frac{1}{2}x - 2 = 1$	$x = 6$
$\frac{1}{2}x - 2 = -1$	$x = 2$

1b) Below is the graph of the equation $y = 5x - 3$. Use it to solve the equations.



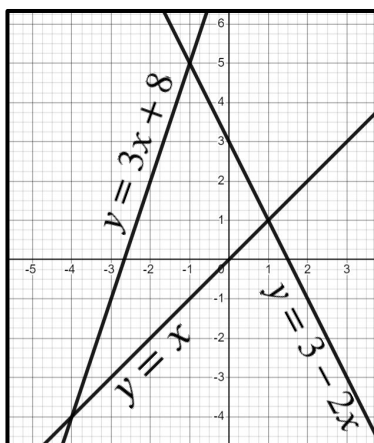
equation	solution
$5x - 3 = -3$	$x = 0$
$5x - 3 = -8$	$x = -1$

1c) Solve the equations below by drawing the relevant graph.



equation	solution
$2x + 1 = 3$	$x = 1$
$2x + 1 = 1$	$x = 0$

2) Use the lines on the graph to solve the equations.



equation	solution
$x = 3 - 2x$	$x = 1$

equation	solution
$x = 3x + 8$	$x = -4$

equation	solution
$3x + 8 = 3 - 2x$	$x = -1$

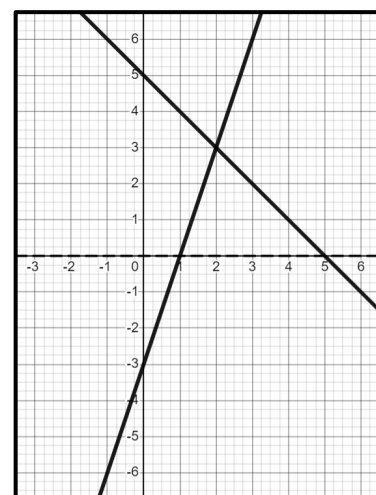
3) Solve the equation below by drawing the relevant graphs.

$$y = 3x - 3$$

x	-1	0	1	2	3
y	-6	-3	0	3	6

$$y = 5 - x$$

x	-1	0	1	2	3
y	6	5	4	3	2



equation	solution
$3x - 3 = 5 - x$	$x = 2$

